

PS-01 — Smart Complaint Management System

Description

Students regularly face issues related to Wi-Fi, classrooms, laboratories, hostels, transport, washrooms, electrical systems, and other campus facilities. However, complaints may be difficult to categorize, prioritize, forward to the appropriate department, and track until resolution.

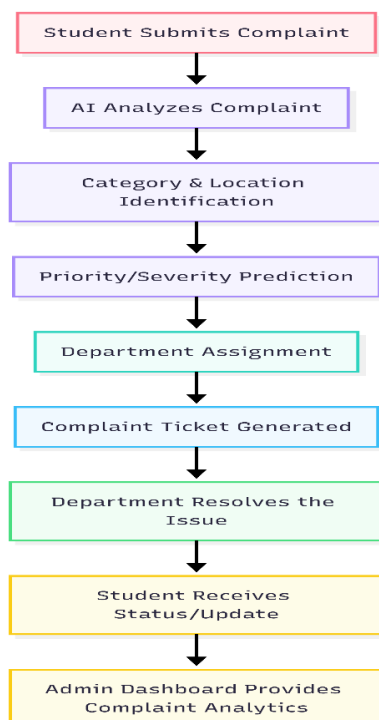
Develop an **AI-powered Smart Complaint Management System** that automatically understands student complaints, identifies the nature and urgency of the issue, and routes it to the appropriate department for timely resolution.

Objectives

Participants are expected to develop a system that:

- Automatically categorizes student complaints.
- Identifies the location and nature of the reported issue.
- Determines the priority/severity of the complaint.
- Routes complaints to the appropriate department.
- Identifies similar or duplicate complaints.
- Enables students to track complaint status.
- Provides administrators with insights into recurring campus issues.

Expected Workflow



Expected Outcomes

- Automated complaint categorization and routing.
- Priority-based complaint management.
- Duplicate/similar complaint identification.
- Centralized complaint tracking for students and administrators.
- Dashboard showing complaint trends, departments, locations, and resolution status.
- Identification of frequently occurring campus problems to support better administrative decisions.
- Improved transparency and faster resolution of student complaints.

PS-02 — Smart Classroom and Seminar Hall Allocation

Description

Educational institutions conduct regular classes, seminars, workshops, guest lectures, examinations, meetings, and other academic activities that require classrooms or seminar halls.

Allocating these spaces manually is challenging because different rooms have different **capacities, facilities, availability, and suitability for specific activities**. At the same time, multiple departments may require the same room or time slot.

This can lead to problems such as **room conflicts, double bookings, assigning a small room to a large group, underutilization of larger rooms, unused rooms during available time slots, and difficulty in finding suitable alternatives when a room is unavailable**.

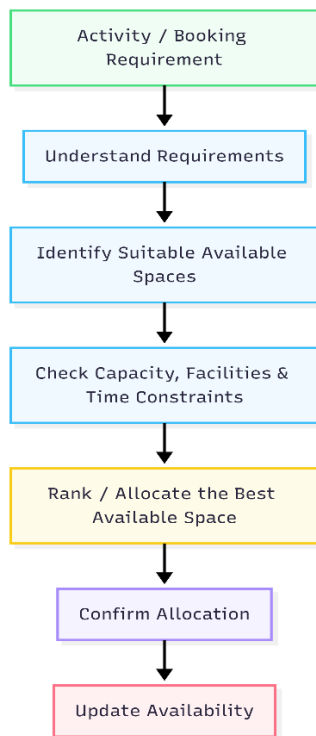
The challenge is to develop an intelligent system that can determine the **most suitable classroom or seminar hall for a given requirement while efficiently utilizing the available spaces and avoiding scheduling conflicts**.

Objectives

Participants are expected to develop a solution that can:

- Identify a suitable classroom or seminar hall based on the requirements of an activity.
- Consider the expected number of participants and room capacity.
- Consider the facilities required for the activity.
- Consider room availability and scheduling constraints.
- Avoid conflicts and double bookings.
- Efficiently utilize available classrooms and seminar halls.
- Provide suitable alternatives when the preferred space is unavailable.
- Support different types of academic and institutional activities.

Expected Workflow



Expected Outcomes

- An intelligent classroom and seminar hall allocation system.
- Conflict-free allocation of available spaces.
- Suitable room recommendations based on activity requirements.
- Better utilization of classrooms and seminar halls.
- Alternative space recommendations when conflicts occur.
- A clear view of room allocation and availability.
- The solution should demonstrate how intelligent decision-making can improve the efficiency of institutional space management.

PS-03 — Campus Classroom Navigator

Description

Students, especially new students and visitors, often face difficulty finding classrooms, laboratories, seminar halls, offices, libraries, and other facilities within a large college campus. Multiple buildings, floors, corridors, entrances, and similarly numbered rooms can make it difficult to identify the correct destination and route.

The challenge is to develop a **smart campus navigation system** that helps a student find the exact location of a required classroom or campus facility and provides a clear route from the student's current location to the destination.

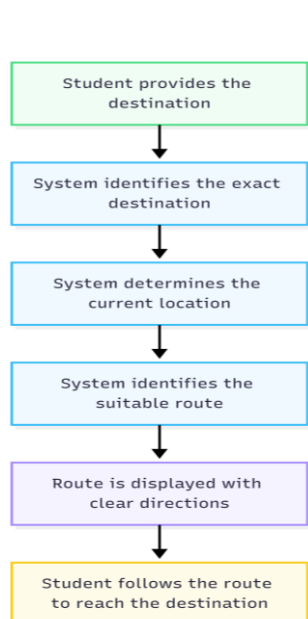
The system should make campus navigation simple and understandable even for someone who is unfamiliar with the campus.

Objectives

Participants are expected to develop a system that can:

- Understand the destination requested by the student.
- Identify the exact classroom, laboratory, seminar hall, office, or other campus facility.
- Determine the student's current location.
- Identify a suitable route from the current location to the destination.
- Provide clear and easy-to-follow navigation instructions.
- Support destinations located in different buildings and floors.
- Suggest an alternative route when the preferred route is unavailable.
- Help new students, visitors, faculty, and staff navigate the campus easily.

Expected Workflow



Expected Outcomes

- A user-friendly campus navigation system.
- Accurate identification of classrooms and campus facilities.
- Clear, step-by-step route guidance.
- Navigation across multiple buildings and floors.
- Suitable or shortest-route recommendations.
- Easier campus navigation for new students and visitors.
- A scalable solution that can support classrooms, laboratories, seminar halls, offices, libraries, hostels, canteens, and other campus facilities.

PS-04 — Intelligent Department Timetable Scheduling

Description

Designing a timetable for a department is a complex task because it must satisfy the requirements of **students, faculty, courses, classrooms, and laboratories** at the same time. Manual timetable preparation can result in faculty clashes, overlapping classes, inefficient room utilization, uneven distribution of classes, and difficulty accommodating faculty or student preferences.

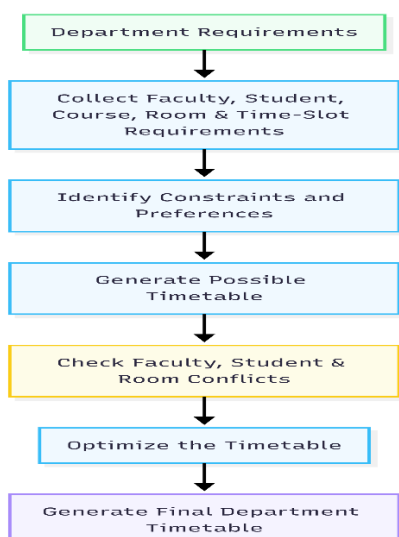
The challenge is to develop an **intelligent timetable scheduling system** that generates an effective department timetable while considering the requirements and constraints of faculty members and students.

Objectives

Participants are expected to develop a system that can:

- Generate a timetable for an entire department.
- Consider faculty availability and preferences.
- Consider student batch and course requirements.
- Avoid faculty and student timetable clashes.
- Allocate classrooms and laboratories appropriately.
- Consider the required number of hours for each subject.
- Distribute classes effectively across available time slots.
- Handle different requirements for theory, laboratory, and elective courses.
- Provide an alternative timetable when the initial schedule cannot satisfy all requirements.

Expected Workflow



Expected Outcomes

- An automatically generated department timetable.
- Conflict-free scheduling of classes.
- Better consideration of faculty and student requirements.
- Efficient utilization of classrooms and laboratories.
- Balanced distribution of classes across the working week.
- Reduced manual effort in timetable preparation.
- Ability to modify and regenerate the timetable when requirements change.
- A clear timetable that can be accessed by students and faculty.

PS-05 — AI-Powered Student Help Desk Chatbot

Description

Students frequently need information about **examinations, attendance, fees, scholarships, certificates, academic procedures, departments, hostel facilities, transport, placements, events, and other college services**. Finding the correct information can require visiting different departments, checking multiple notices, or asking faculty and administrative staff.

The challenge is to develop an **AI-powered Student Help Desk Chatbot** that allows students to ask their questions in natural language and receive **clear, relevant, and reliable information** about college-related services and procedures.

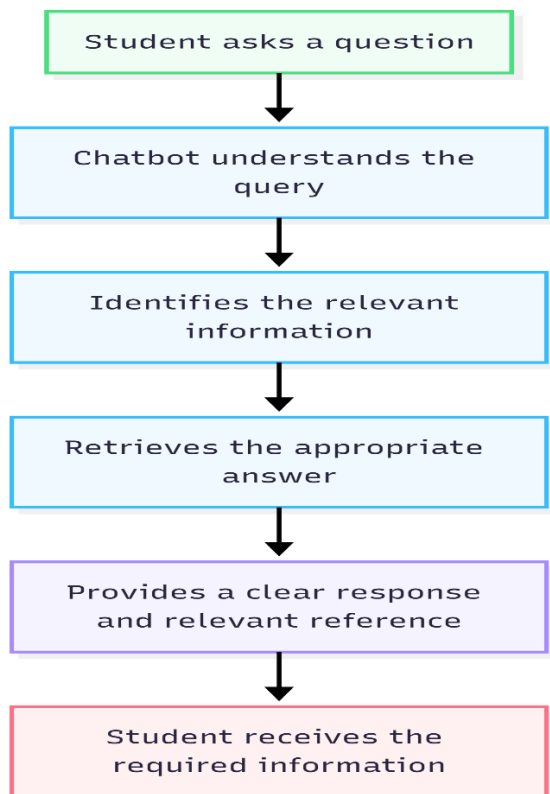
The chatbot should be designed to assist students at any time and reduce the need for students to approach multiple departments for routine queries.

Objectives

Participants are expected to develop a system that can:

- Understand student questions expressed in natural language.
- Provide relevant answers to common college-related queries.
- Handle queries related to academics, examinations, fees, attendance, certificates, scholarships, hostel, transport, placements, and events.
- Guide students through college procedures and required steps.
- Provide the relevant source or reference for important information.
- Handle questions that are unclear or outside the available information appropriately.
- Provide a simple and user-friendly interface for students.

Expected Workflow



Expected Outcomes

- A user-friendly AI-based student help desk.
- Quick access to college-related information.
- Natural-language interaction between students and the system.
- Reduced dependency on multiple departments for routine queries.
- Clear guidance for college procedures and services.
- Source-backed and reliable responses.
- A system that can be extended to support different departments and student services.

PS-06 — Smart Lost and Found Management System

Description

Students, faculty, and staff frequently lose or find personal and academic belongings such as **ID cards, wallets, mobile phones, books, certificates, keys, bags, electronic devices, and other items** within the college campus. Currently, identifying the owner of a found item or locating a lost item can be difficult because there is no centralized system to connect lost and found reports.

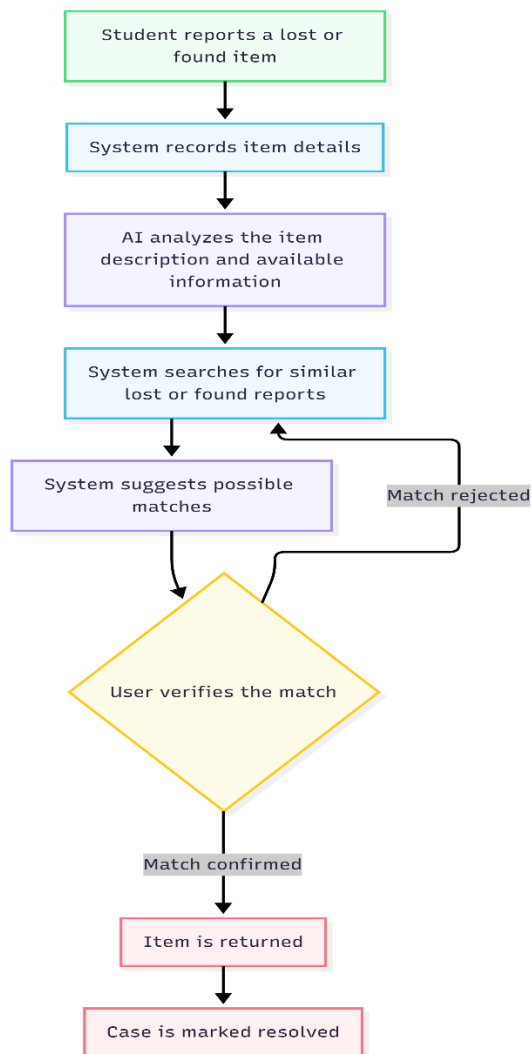
The challenge is to develop a **Smart Lost and Found Management System** that helps users report lost or found items and intelligently identify possible matches between them.

Objectives

Participants are expected to develop a system that can:

- Allow users to report lost or found items.
- Capture important details such as item type, description, location, and date.
- Identify similarities between lost and found item reports.
- Suggest possible matches to the user.
- Help verify whether a found item belongs to a particular person.
- Track the status of reported items.
- Provide a centralized platform for managing lost and found items.

Expected Workflow



Expected Outcomes

- Centralized college lost and found platform.
- Faster identification of lost belongings.
- Intelligent matching of lost and found items.
- Possible-match recommendations for users.
- Item status and case tracking.
- Reduced manual effort in managing lost and found items.
- Improved chances of returning lost belongings to their rightful owners.

PS-07 — Exam Seating Arrangement Optimizer

Description

During examinations, students from different classes or sections need to be assigned seats across multiple examination rooms. Preparing the seating arrangement manually can be time-consuming and may result in **student clashes, incorrect room allocation, uneven room utilization, and difficulty maintaining appropriate separation between students from the same class or section.**

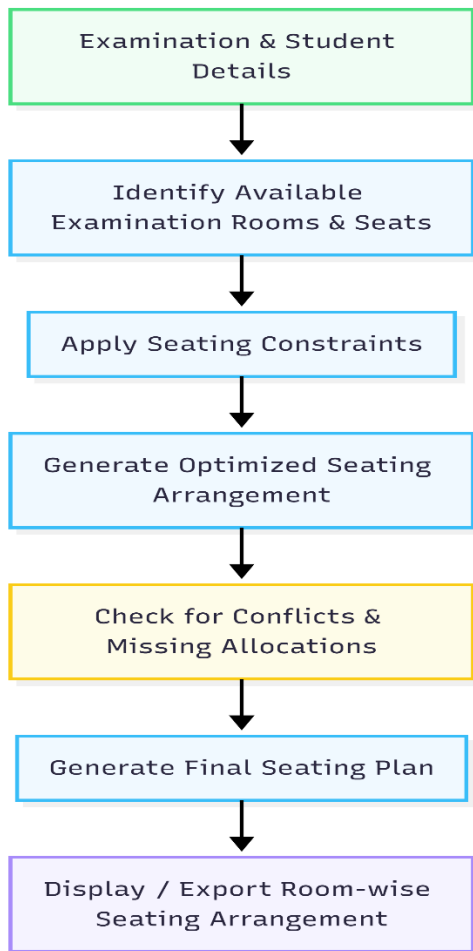
The challenge is to develop an **intelligent exam seating arrangement system** that automatically generates an optimized seating plan while considering examination rooms, available seats, student groups, and examination requirements.

Objectives

Participants are expected to develop a system that can:

- Generate seating arrangements for students across available examination rooms.
- Allocate students to appropriate rooms based on available seating capacity.
- Avoid placing students from the same class or section together where required.
- Prevent duplicate or missing student allocations.
- Ensure that every eligible student receives exactly one seat.
- Make efficient use of available examination rooms and seats.
- Handle different examination sessions and student groups.
- Allow the seating arrangement to be regenerated when room availability or student information changes.

Expected Workflow



Expected Outcomes

- Automated exam seating arrangement.
- Conflict-free allocation of students to seats.
- Proper distribution of students across examination rooms.
- Efficient utilization of available seats.
- Prevention of duplicate or missing student assignments.
- Room-wise and student-wise seating information.
- Ability to regenerate seating arrangements when requirements change.
- A clear seating plan that can be easily used by examination staff and students.

PS-08 — Intelligent Faculty Workload Balancing System

Description

Faculty members are assigned different academic and administrative responsibilities such as **teaching classes, laboratory sessions, project guidance, examinations, mentoring, departmental activities, and other institutional duties**. When these tasks are allocated manually, the workload may become uneven, with some faculty members being **overloaded** while others remain **underutilized**.

The challenge is to develop an **intelligent faculty workload management system** that analyzes the tasks assigned to faculty members and ensures that their workload is distributed appropriately according to their **designation and prescribed workload norms**.

For example, the system should consider the applicable workload limits for different faculty categories:

- **Professor:** 10 hours
- **Associate Professor:** 16 hours
- **Assistant Professor:** 20 hours

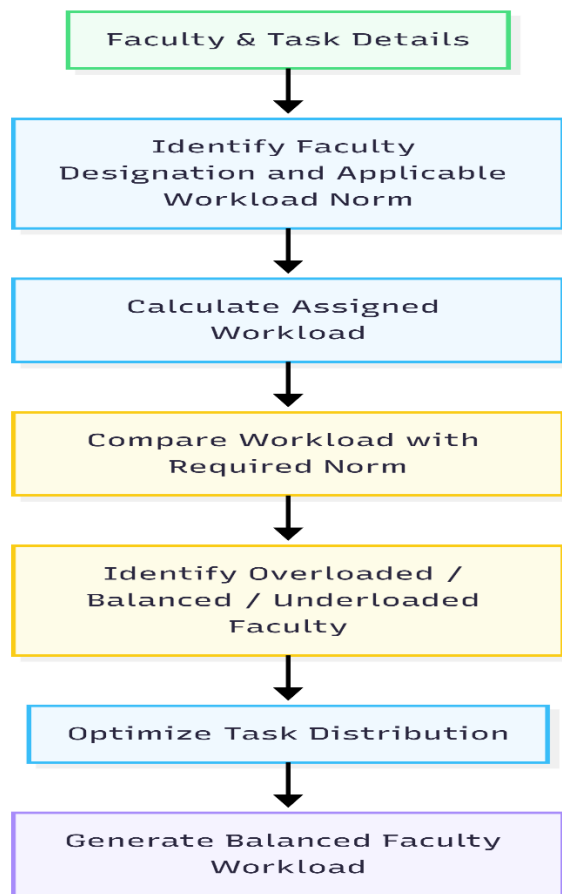
The system should identify workload imbalances and help generate a more balanced allocation of responsibilities.

Objectives

Participants are expected to develop a system that can:

- Calculate the total workload assigned to each faculty member.
- Consider different workload limits based on faculty designation.
- Identify faculty members who are **underloaded, appropriately loaded, or overloaded**.
- Consider different types of academic and institutional tasks.
- Identify workload imbalances within the department.
- Recommend a more balanced distribution of available tasks.
- Ensure that reallocating tasks does not create conflicts with faculty availability or academic requirements.

Expected Workflow



Expected Outcomes

- Faculty-wise workload calculation.
- Identification of overloaded and underloaded faculty members.
- Workload comparison based on faculty designation.
- Balanced distribution of teaching and other assigned responsibilities.
- Reduced workload imbalance within the department.
- Clear workload summary for faculty and department administrators.
- Ability to modify task assignments and regenerate an optimized workload distribution.